

Spencer Yael Weinstein

PhD Candidate
Department of Biology, University of Waterloo
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she/her

Education

- 2023 (expected) **PhD in Biology**, University of Waterloo
Dissertation: Diversity of chars in the Coppermine River near Kugluktuk, Nunavut
Advisor: Dr. Heidi Swanson
- 2018 **M.S. in Biology**, Texas Christian University
Thesis: Should I stay or should I go? Analyzing the genetic basis of migration-related traits in rainbow trout (*Oncorhynchus mykiss*)
Advisor: Dr. Matthew Hale
- 2016 **B.S. in Natural Resources Conservation**, University of Massachusetts, Amherst
B.A. in Geography, University of Massachusetts, Amherst
Thesis: Genetic analysis of slimy sculpin (*Cottus cognatus*) movement before and after dam removal
Advisors: Drs. Andrew Whiteley and Allison Roy
Commonwealth Honors College, Multidisciplinary Honors with Greatest Distinction
Five College Certificate in Coastal and Marine Sciences
Summa cum laude

Research Experience (Non-Thesis)

- 2017 **Graduate Research Assistant**, Texas Christian University
Mercury risk to fish in the south-central U.S.: effects of mercury deposition, fish trophic position, and fish length.
Advisors: Drs. Ray Drenner and Matthew Chumchal
- 2015 **NSF REU Scholar**, Virginia Institute of Marine Science
Characterization of blueline tilefish (*Caulolatilus microps*) microsatellite loci: a genetic investigation of geographic connectivity
Advisor: Dr. Jan McDowell
- 2014 **NSF REU Scholar**, South Dakota State University
Identification of mitogen-activated protein kinases in prairie cordgrass (*Spartina pectinata*)
Advisors: Drs. Madhav Nepal and Jose Gonzalez-Hernandez

Publications

- 2019 **Weinstein SY**, Thrower FP, Nichols KM, Hale MC. 2019. *A large-scale chromosomal inversion is not associated with life history development in rainbow trout from Southeast Alaska*. PLoS ONE, 14(9): e0223018.
- 2019 **Weinstein SY**, Coombs JA, Nislow KH, Riley C, Roy AH, Whiteley AR. 2019. *Evaluating the effects of barriers on slimy sculpin movement and population connectivity using novel sibship-based and traditional genetic metrics*. Transactions of the American Fisheries Society, 148(6), 1117-1131.

Awards, Grants, and Honors

2022	ACUNS/APECS Canada Illustrate Your Research Contest
2022	William R. Mote Fisheries Fellowship Award
2022	American Fisheries Society – International Fisheries Section (AFS-IFS) Fellow
2021	James E. Wright Graduate Award
2021	Adrian Dahood-Fritz Memorial Fund – Small Research Grant
2021	John E. Skinner Memorial Award
2020	Vanier Canada Graduate Scholarship
2020	University of Waterloo President's Graduate Scholarship
2019	Davis Memorial Scholarship in Ecology
2019	Royal Bank of Canada Water Scholarship
2018	Coastwide Salmonid Genetics Meeting Travel Award
2017	Adkins Fellowship for Summer Research
2016	UMass Outstanding Senior in Geography
2016	UMass Outstanding Senior in Fisheries Conservation
2016	Best Oral Presentation in Ecology, Environmental, and Earth Sciences, Emerging Researchers National Conference in STEM
2014	First Place, South Dakota State University Bioenergy REU Poster Symposium
2014	Esther Warner Natural Resources Conservation Scholarship
2014	AT&T Scholarship
2012	UMass Chancellor's Scholarship

Outreach and Leadership

2022 –	NoWPaS 2023 Organizing Committee
2021 – 2022	National Relations Coordinator, APECS Canada
2021 – 2022	President, University of Waterloo Biology Graduate Student Association
2021 – 2022	NoWPaS 2022 Organizing Committee
2020 – 2022	American Fisheries Society Climate Ambassador
2020 – 2021	Council Member, Association of Polar Early Career Scientists
2020 – 2021	NoWPaS 2021 Organizing Committee
2020 – 2021	Vice President, University of Waterloo Biology Graduate Student Association
2019 – 2020	Conference Coordinator, University of Waterloo Biology Graduate Student Association
2019	Co-coordinator, Kitchener-Waterloo 500 Women Scientists Pod
2019	Workshop Facilitator, TD Walter Bean Science Outreach Day
2018 –	Participant, Skype a Scientist
2017 –	Pen-pal, Letters to a Pre-Scientist

Workshops and Training

2021	Physalia Courses – Geometric Morphometrics
2021	Sentinel North Indigenous Engagement Workshop
2020	ComSciCon Greater Toronto Area
2020	Diversity, Equity, and Inclusion in Socio-Environmental Synthesis Workshop, National Socio-Environmental Synthesis Center (SESYNC)
2020	Summer Institute in Statistical Genetics

Teaching Experience

2020	Teaching Assistant, Methods in Bioinformatics (Biol 365), University of Waterloo
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2019	Teaching Assistant, Computational Biology (Biol 266), University of Waterloo
2017, 2018	Teaching Assistant, Introductory Biology II (Biol 10511), Texas Christian University
2017	Teaching Assistant, Genomics (Biol 40274), Texas Christian University
2016, 2017	Teaching Assistant, Contemporary Issues in Biology (Biol 10003), Texas Christian University

Presentations

2022	Weinstein SY* , Hale MC, Loewen TN, Swanson HK. <i>A mixed methods approach to characterizing the diversity of chars (Salvelinus spp.) in the central Canadian Arctic</i> . Fisheries Society of the British Isles Annual Symposium. Nottingham Trent University, UK. *Attended as 2022 AFS Traveling Fellow.
2022	Weinstein SY , Hale MC, Loewen TN, Swanson HK. <i>A mixed methods approach to characterizing the diversity of chars (Salvelinus spp.) in the central Canadian Arctic</i> . NoWPaS: International Workshop for PhD Students on Anadromous Salmonids. Basque Country, France.
2021	Weinstein SY , Hale MC, Loewen T, Swanson HK. [Poster] <i>Diversity of chars in the central Canadian Arctic</i> . Ontario Chapter of the American Fisheries Society. Virtual.
2021	Weinstein SY , Hale MC, Loewen T, Swanson HK. <i>Diversity of chars in the central Canadian Arctic</i> . NoWPaS: International Workshop for PhD Students on Anadromous Salmonids. Virtual.
2020	Weinstein SY , Hale MC, Swanson HK. <i>Diversity of chars in the Coppermine River near Kugluktuk, Nunavut</i> . American Fisheries Society Annual Virtual Meeting.
2020	Weinstein SY , Hale MC, Swanson HK. <i>Diversity of chars in the Coppermine River near Kugluktuk, Nunavut</i> . NoWPaS: International Workshop for PhD Students on Anadromous Salmonids. Laugarvatn, Iceland.
2018	Weinstein SY , Thrower FP, Nichols KM, Hale MC. <i>Should I stay or should I go? Analyzing the genetic basis of migration-related traits in rainbow trout (Oncorhynchus mykiss)</i> . Coastwide Salmonid Genetics Meeting, Mukilteo, Washington.
2018	Weinstein SY , Thrower FP, Nichols KM, Hale MC. <i>Should I stay or should I go? Analyzing the genetic basis of migration-related traits in rainbow trout (Oncorhynchus mykiss)</i> . Texas Chapter of the American Fisheries Society. College Station, Texas.
2017	Weinstein SY , Thrower FP, Nichols KM, Hale MC. [Poster] <i>Should I stay or should I go? Analyzing the genetic basis of migration-related traits in rainbow trout (Oncorhynchus mykiss)</i> . Evolution Meeting. Portland, Oregon.
2016	Weinstein SY , Roy AH, Whiteley AR. <i>Slimy sculpin movement before and after dam removal</i> . Northeast Natural History Conference. Springfield, Massachusetts.
2016	Weinstein SY , Roy AH, Whiteley AR. [Poster] <i>Genetic analysis of slimy sculpin movement before and after dam removal</i> . Massachusetts Statewide Undergraduate Research Conference. Amherst, Massachusetts.
2016	Weinstein SY , Brightman H, McDowell JR. [Poster] <i>Characterization of blueline tilefish (Caulolatilus microps) microsatellite loci: a genetic investigation of geographic connectivity</i> . Five College Coastal and Marine Sciences Spring Poster Symposium. Amherst, Massachusetts.

- 2016 **Weinstein SY**, Brightman H, McDowell JR. *Characterization of blueline tilefish (Caulolatilus microps) microsatellite loci: a genetic investigation of geographic connectivity*. Emerging Researchers National Conference in STEM. Washington, DC.
- 2015 **Weinstein SY**, Foley M, Benson BV, Nepal MP, Gonzalez-Hernandez JL. [Poster] *Identification of mitogen-activated protein kinases in prairie cordgrass (Spartina pectinata)*. Emerging Researchers National Conference in STEM. Washington, DC.
- 2014 **Weinstein SY**, Foley M, Benson BV, Nepal MP, Gonzalez-Hernandez JL. [Poster] *Analysis and identification of mitogen-activated protein kinases in prairie cordgrass (Spartina pectinata)*. SD-EPSCoR South Dakota Undergraduate Research Symposium. Pierre, South Dakota.